Numerical Methods for Differential Equations in Model Rocket Simulation

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Abstract

Since the dawn of the space age, numerical methods have played a defining role in rocket development by enabling engineers to model trajectories long before physical testing. Because many governing equations in flight dynamics lack closed-form solutions (Karbon, 1998), numerical methods are the practical way to analyze and predict behavior. Building on these principles, this project applied those same numerical techniques to the study of small-scale rocket flight, extending a previously verified 1D ascent simulator into a full 3D Python-based flight simulation with explicit geometry, event-aware dynamics, and atmospheric modeling.

The rocket's geometry is defined component-wise (nose, body tubes, transitions, and fins) with full parametric dimensions, from which reference and wetted areas are computed. Atmospheric conditions follow the International Standard Atmosphere (ISA) temperature and pressure model, enabling Mach and Reynolds number corrections, while angle of attack is computed against a configurable wind model, validated here at zero wind to match the actual launch conditions. Aerodynamic forces follow a Barrowman-derived approach for normal force and center of pressure, including nose pressure drag, fin and body contributions, base drag, and transonic corrections consistent with model-rocket practice (Niskanen, 2013). Three-dimensional attitude is represented using quaternions, which rotate aerodynamic and thrust vectors between the body and inertial frames, and translational and rotational dynamics are integrated using a fixed-stage fourth-order Runge-Kutta (RK4) scheme combined with adaptive step-size control that shrinks the timestep around ignition, burnout, and other rapidly changing force events and relaxes it during smoother coasting. This scheme was chosen over the Euler and classical predictor-corrector methods used earlier in the project for its better handling of discontinuities and rapidly changing forces during ascent (Derrick and Grossman, 1997; Bryan, 2025).

The prior 1D model developed for this project reached apogee within 2.00% of OpenRocket's prediction for the same rocket and motor, with its three integrators (Euler, Adams-Bashforth-Moulton, and RK4) agreeing with each other to within 0.1%; this motivated its use as a baseline benchmark as the simulator expanded into 3D flight with attitude dynamics and off-axis forces, and it has since been retired from the codebase now that the 3D model is independently validated. The completed 3D model was validated against OpenRocket's own simulation of the identical rocket, motor, and launch conditions, and against flight-computer data recorded from an actual launch of the physical rocket. Apogee altitude, apogee time, and peak velocity agreed with OpenRocket to within 0.01%, roughly a 200-fold reduction in error relative to the 1D baseline. Comparison against the real flight showed a larger gap, with the real flight's recorded apogee approximately 12.42% higher than either simulation. Because the two independently built simulators agreed with each other far too closely for a shared modeling error to explain a gap that size, this was traced instead to the real flight's own motor and sensor variance: the flight computer's barometric altimeter had no onboard accelerometer, and its raw log required correcting a ground-elevation error, removing duplicate samples, and re-deriving velocity and acceleration from the recorded pressure data with a Kalman filter before a fair comparison against either simulation was possible.

Beyond its computational contribution, this project shows how undergraduate students can apply numerical methods for differential equations to build and validate a complete engineering simulation against both a reference simulator and a real physical flight, turning abstract mathematics into a practical, physically tested tool and forming a foundation for future student research in rocket flight dynamics. In undergraduate engineering, numerical modeling has become especially important because it gives students an accessible means to explore complex real-world systems using mathematics and computation (Niskanen, 2013).

Keywords: numerical methods, ordinary differential equations, Runge-Kutta method, rocket flight simulation, model validation, Kalman filter

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